# End-to-End Evaluation Report — IISc Grounded Agentic RAG Platform

- **Date:** 2026-06-21
- **Branch:** `codex/evaluation` (already contains all of `dev` ; `dev` is 0 commits ahead — the Step-1 merge is a no-op)
- **System under test: live** n8n retrieval webhook — `https://n8n-production-c637.up.railway.app/webhook/retrieve`
- **Transport:** direct-to-n8n (bypasses the FastAPI backend), tenant `iisc-demo` ( `e501e9e6-…39f5` )
- **Harness:** `evaluation/` 148-case / 13-application suite + pinned **RAGAS 0.1.10** independent judge (OpenAI)
- **Result bundle:** `artifacts/evaluation_results/n8n_endpoint_eval_20260621T2029.{json,csv,md,html}`
- **Companion:** `TEST_COVERAGE_AND_FEATURE_GAPS_20260621T2029.md` (Step 4 — what else to test + per-feature gaps)
- **Supersedes context from:** `M6_n8n_evaluation_report.md` (2026-06-20), in which the live eval could **not** be run.

> *This is the **first true live RAGAS evaluation** of the deployed retrieval engine. The prior M6 report graded the workflow statically and explicitly marked the live track "NOT RUN."*

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## 1. Executive Summary

| Track | Result |
|---|---|
| Transport / availability | ✅ **148/148 calls succeeded, 0 errors.** Endpoint live and stable. |
| Retrieval quality | ✅ **Strong.** context_precision **0.979**, context_recall **0.923**, citation coverage **1.00**. |
| Abstention on unanswerable | ✅ **Perfect — 16/16** correctly refused (see §4; the harness's 0.0 is a detector bug). |
| Generation grounding | ❌ **Below bar.** faithfulness **0.830** < 0.85 gate; concentrated in technical domains. |
| Answer relevancy | ✅ **0.953.** |
| **RAGAS quality gate** | ❌ **FAILS on faithfulness only** (3 of 4 metrics pass). |
| Latency | ❌ **Unacceptable for interactive use** — mean **43.3 s**, p95 **49.7 s**, max **58.7 s**. |
| Source-of-truth integrity | ❌ **Production ≠ repository** (see §5.1 — the grounding fix is on an unmerged branch). |

**Bottom line.** The deployed retrieval engine **works and retrieves well** — it pulls the right chunks, cites them, and correctly abstains when the answer isn't in the corpus. Two things hold it back from production-grade: (1) **generation faithfulness** — the model adds claims beyond the retrieved context in technical domains; and (2) **latency** — ~43 s per query makes the WhatsApp/chat experience untenable. Separately, there is a **governance problem**: the version of the workflow that produced these (mostly good) numbers **is not the one committed to the repo**.

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## 2. What Was Run

Every suite question was POSTed straight to the live n8n webhook with the backend's exact payload contract ( `{request_id, tenant_id, conversation_id, current_message, history}` ), and each answer + retrieved chunks were graded by an independent RAGAS judge. No mock, no backend in the path.

- 148 cases · 13 applications · 132 answerable · 16 unanswerable.
- Concurrency 6, 0 transport errors, every case returned exactly 3 retrieved chunks.
- Independent judge (RAGAS 0.1.10, era-correct stack) scored the 132 answerable cases.

*Reproducing this was non-trivial:* the live host's name fails Python's `getaddrinfo` in this environment (works via `dig` / `host` ), so the runner pins the host IP; RAGAS' own event loop and a `datasets` dtype edge had to be worked around. The runner is `scratchpad/run_eval_n8n_endpoint.py` .

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## 3. Live RAGAS Scores

| Metric | Score | Gate | Verdict |
|---|---|---|---|
| Faithfulness | **0.830** | ≥ 0.85 | ❌ **FAIL** |
| Answer relevancy | **0.953** | ≥ 0.80 | ✅ |
| Context precision | **0.979** | ≥ 0.75 | ✅ |
| Context recall | **0.923** | ≥ 0.80 | ✅ (computed on 121/132; 11 RAGAS judge timeouts → excluded) |
| Citation coverage | **1.000** | — | ✅ |
| Negative abstention (true) | **1.000 (16/16)** | — | ✅ (harness reported 0.0 — see §4) |

**Latency (seconds):** mean **43.3** · p50 **42.1** · p95 **49.7** · max **58.7**.
**Generator used:** `openai` on **100% (148/148)** of responses — *not* the documented DeepSeek-primary + Gemini-self-check chain.

### 3.1 Faithfulness by application (worst first) — *where to improve*

| Application | n | Faithfulness |
|---|---|---|
| kubernetes_troubleshooting | 28 | **0.65** |
| incident_response | 6 | **0.67** |
| customer_support | 7 | **0.69** |
| it_helpdesk | 7 | 0.76 |
| healthcare_administration | 7 | 0.80 |
| education_assistant | 7 | 0.86 |
| legal_document_navigation | 7 | 0.88 |
| compliance_policy | 7 | 0.89 |
| sales_enablement | 7 | 0.90 |
| employee_onboarding | 7 | 0.93 |
| bug_reporting | 27 | 0.94 |
| developer_documentation | 7 | 1.00 |
| equipment_maintenance | 7 | 1.00 |

**Reading:** retrieval is uniformly strong (precision 0.98, recall 0.92), so the faithfulness shortfall is a **generation** problem, not a retrieval one. It clusters in domains where the base model has strong priors (Kubernetes, incident response, IT/support): the `openai` generator answers partly from its own training rather than strictly from the retrieved chunks, so RAGAS flags ungrounded claims. Domains with corpus-specific content (equipment, dev-docs, bug-reporting) score 0.94–1.00 because the model has no priors to drift toward.

## 4. The Abstention Result (and a harness bug)

All **16/16** unanswerable questions were correctly refused. Examples:

- "*Which Helm chart version fixes the documented CoreDNS timeout issue?*" →
  "The document does not provide information about which Helm chart version fixes…"
- "*What is the monthly price of a larger managed Kubernetes node pool?*" →
  "The provided context does not include information regarding the monthly price…"

The summary's `negative_abstention_rate: 0.0` is **wrong** — a harness defect, not a system defect. `run_eval.build_summary` detects abstention via (a) `requires_clarification=true` (never returned), (b) empty contexts (never empty — retrieval always returns 3), or (c) the literal phrases *"do not contain"* / *"don't have enough information."* The model abstains with different wording (*"does not provide information"*, *"does not include information"*), so the detector misses all 16. **Fix the detector** (broaden the phrase set / add an LLM-judged abstention check) — tracked in the companion report. Real-world abstention behavior here is excellent.

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## 5. Areas of Improvement (prioritized)

### P0 — Critical

**5.1 Production runs code that isn't in the repo (source-of-truth drift).**
The committed `n8n-workflows/retrieval-pipeline.json` on `dev` / `codex/evaluation` still has **all** the original M6 defects: **no `tenant_id` filter** (0 occurrences) and it reads **`body.query`** (4×) instead of `current_message`. The working, grounded, tenant-scoped version that produced these results exists only on the **unmerged** branch `fix/m6-retrieval-grounding` (commit `59422f2`, "ground retrieval pipeline — pass query, scope by tenant, return sources"). So: **the live endpoint is correct, but anyone deploying from `dev` would ship the broken, cross-tenant-leaking workflow.**
*→ Merge `fix/m6-retrieval-grounding` into `dev` and re-export the deployed workflow JSON. Treat the n8n export as a reviewed, versioned artifact.*

**5.2 Latency is ~43 s/query (p95 ~50 s).** Unusable for chat/WhatsApp and far over any interactive budget. Driver: the 100%-`openai` generation path plus the embed→search→rerank→generate(→self-check) chain. → *Profile per-node latency; cut the fallback fan-out on the happy path; stream tokens; consider a faster primary generator and parallelizing rerank. Target p95 < 8–10 s.*

### P1 — High

**5.3 Faithfulness 0.83 < 0.85, concentrated in technical domains (§3.1).** The generator supplements retrieved context with world knowledge. → *Tighten the system prompt to "answer ONLY from context; if unsupported, say so"; add/repair the Gemini self-check to reject ungrounded claims; re-evaluate generator choice (the earlier deepseek path scored 0.86 faithfulness — see §6).*

**5.4 Response breaks the documented API contract & is unstable.** The live `sources` schema is `{id, content}`. The OpenAPI `SourceChunk` promises `{document_id, title, chunk_text, page_number, score}` — so `title`, `page_number`, `score`, and a stable `document_id` are **missing**, and the field even **changed mid-evaluation** (`{chunk_text,title}` → `{id,content}`). The frontend's citations panel and the `faithfulness` / `requires_clarification` / `follow_up_questions` fields the UI expects are **not** returned. → *Conform the n8n "Respond to Webhook" node to the spec; add a contract test on both sides (companion report G2).*

**5.5 Tenant isolation is correct in prod but unguarded.** Verified working live, but there is **no automated test** and the committed workflow drops the filter entirely (5.1). One bad re-export re-introduces cross-tenant leakage silently. → *Add the cross-tenant isolation test (companion G1).*

### P2 — Medium

- **5.6 RAGAS judge timeouts (11/132 recall, 2 relevancy, 1 faithfulness).** Driven by the ~43 s system latency colliding with judge timeouts. Lower system latency (5.2) and/or raise judge timeout + concurrency so recall is computed on the full set.
- **5.7 Grounding signals absent.** No `faithfulness` / `requires_clarification` / `follow_up_questions` in the response, so the UI badges/clarification banner can't function end-to-end.
- **5.8 Ephemeral-upload retrieval not in the main path.** Main retrieval searches only `document_chunks`; WhatsApp/Teams/chat file uploads land in `ephemeral_chunks` and aren't queried — users can't ask about a file they just sent.
- **5.9 Deployment artifacts incomplete.** `frontend/Dockerfile` is **missing** (compose build fails); onboarding endpoints are `501` stubs; several admin pages are mock placeholders.

### P3 — Lower

- **5.10 Secrets default to `change-me-…`** for `SECRET_KEY`, `N8N_CALLBACK_TOKEN`, `MCP_API_KEY` — must be rotated per environment.
- **5.11 Observability:** stdout logs + `/health` only; no structured logs, request timing, metrics, or n8n/frontend health checks.
- **5.12 Doc drift:** `PROJECT_SPEC.md` still names OpenAI `text-embedding-3-small` + `gpt-4o-mini`; the system uses Voyage `voyage-4-large` + (now) an `openai` generator. The deployed generator also contradicts the documented DeepSeek-primary design.

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## 6. Comparison Across Runs

| Metric | Prior M6 report (06-20) | Same-day backend run `…055215Z` | This live endpoint run |
|---|---|---|---|
| Live eval performed? | ❌ not run (no stack) | ✅ via backend `/chat/query` | ✅ direct to n8n webhook |
| Faithfulness | — | 0.860 ✅ | **0.830** ❌ |
| Answer relevancy | — | **0.667** ❌ | **0.953** ✅ |
| Context precision | — | 0.934 ✅ | 0.979 ✅ |
| Context recall | — | 1.000 ✅ | 0.923 ✅ |
| Citation coverage | — (predicted 0) | 1.000 | 1.000 |
| Generator | deepseek (designed) | deepseek | **openai (100%)** |
| Gate | ❌ (static defects) | ❌ (relevancy) | ❌ (faithfulness) |

**Insight:** the generator changed between the two same-day runs (deepseek → openai). That swap **fixed answer relevancy** (0.667 → 0.953) but **cost faithfulness** (0.86 → 0.83) — openai produces more fluent, on-point answers but drifts from the source on technical topics. The prior M6 report's core prediction held: without the (now-applied-in-prod) fixes, the pipeline returns no usable citations. Those fixes work — they're just **not yet merged** (5.1).

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## 7. Recommended Action Plan

1. **Merge `fix/m6-retrieval-grounding` → `dev`** and re-export the live workflow JSON into the repo (5.1). *Blocker for any honest deploy.*
2. **Attack latency** to p95 < 10 s (5.2) — biggest UX lever.
3. **Raise faithfulness ≥ 0.85** via prompt hardening + a working self-check; re-benchmark deepseek vs openai on the faithfulness/relevancy trade-off (5.3, §6).
4. **Conform the webhook response to the OpenAPI `SourceChunk` contract** and add two-sided contract tests (5.4).
5. **Land the cross-tenant isolation test + abstention-detector fix** (5.5, §4; companion G1/G3).
6. **Re-run this exact harness in CI nightly** with `--enforce-thresholds`; publish the HTML bundle and alert on regression.

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## 8. References

- Live result bundle: `artifacts/evaluation_results/n8n_endpoint_eval_20260621T2029.{json,md,html}`
- Runner: `scratchpad/run_eval_n8n_endpoint.py` (direct-to-n8n collector + RAGAS scorer)
- Harness contract: `evaluation/run_eval.py`; dataset: `evaluation/application_suite.jsonl` (148 cases)
- Prior static report: `artifacts/evaluation_results/M6_n8n_evaluation_report.md`
- Grounding fix (unmerged): branch `fix/m6-retrieval-grounding` @ `59422f2`
- Companion: `artifacts/evaluation_results/TEST_COVERAGE_AND_FEATURE_GAPS_20260621T2029.md`